

ASSOCIATIONS BETWEEN SLEEP AND MUSCLE HEALTH ACROSS THE LIFESPAN: THE INFLUENCE OF SEX AND AGE

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INTRODUCTION

Skeletal muscle health is an important determinant of physical function, metabolic health and healthy ageing [1]. Muscle mass, strength and physical function progressively decline across the lifespan, although the magnitude and trajectory of these changes differ between males and females [1,2]. Sleep represents a potentially modifiable factor that may influence muscle health, with inadequate or poor-quality sleep associated with lower muscle mass, reduced muscular strength and an increased risk of sarcopenia [3,4]. Despite growing evidence linking sleep and skeletal muscle health, it remains unclear whether these relationships differ according to sex and age. This study therefore examines associations between objectively measured sleep characteristics and markers of skeletal muscle health in males and females across the adult lifespan, while also exploring relationships with circulating sex hormones.

METHODS

This study is a secondary analysis of data collected from the previously completed FAME and MALE studies [5] (DUHREC 2021-307), comprising 178 adult females and males across the lifespan. Sleep was objectively assessed using Actical wrist accelerometry (MiniMitter, Philips Respironics, Bend, OR) and analysed using the Actiware software, with sleep periods verified against participant-completed sleep diaries. Objectively measured sleep variables include total sleep time, sleep efficiency, sleep onset, sleep offset, sleep onset latency, wake after sleep onset and sleep fragmentation. Muscle-health variables include lean muscle cross-sectional area, quadriceps cross-sectional area, body fat percentage, appendicular lean mass and estimated one-repetition maximum strength. Female sex-hormone variables include progesterone (P4), oestradiol (E2), follicle-stimulating hormone (FSH), luteinising hormone (LH), testosterone (T), free oestrogen index (FEI), free androgen index (FAI) and the testosterone-to-oestradiol (T:E2) ratio. Male sex-hormone variables include testosterone, E2, dihydrotestosterone (DHT), FEI, FAI and the T:E2 ratio. Statistical analyses are being conducted in GraphPad Prism using correlation analyses to examine relationships between sleep

characteristics and muscle-health variables, as well as relationships with sex hormones. Statistical significance will be set at $p < 0.05$. Sex- and age-specific interactions will also be explored.

RESULTS & DISCUSSION

Following analysis and verification of the sleep data via sleep diaries, the cohort comprised 75 females and 38 males between the ages of 18-80 years. Among the 75 females with available menopausal-status data, 37 (49.3%) were pre-menopausal, 4 (5.3%) were perimenopausal and 34 (45.3%) were post-menopausal. Statistical analysis of the sleep and muscle-health variables is currently ongoing.

CONCLUSION

It is anticipated that poorer sleep characteristics, including lower sleep efficiency, greater sleep fragmentation, longer sleep onset latency and altered sleep duration or timing, will be negatively associated with markers of muscle health, including lower lean mass and muscular strength. Associations between sleep characteristics and circulating sex hormones are expected to differ between males and females and across age. These findings may improve understanding of the relationship between sleep, hormonal profiles and muscle health across the adult lifespan and help identify sleep as a potentially modifiable factor relevant to the maintenance of muscle health.

ACKNOWLEDGEMENTS

The authors acknowledge Annabel J. Critchlow and Ross M. Williams for their contributions to data collection. This study was supported by an Australian Research Council (ARC) Future Fellowship awarded to Professor Séverine Lamon (FT210100278).

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